Blake Bullwinkel

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Summary

AI safety and security researcher studying vulnerabilities in LLMs and agentic systems including jailbreaks, prompt injection, and model poisoning. Experience in automated red teaming, RL-based model (un)alignment, and backdoor detection. Led red teaming of frontier AI systems, translating research insights into novel attacks and defenses. Coauthor of multiple AI safety/security papers and contributor to Microsoft's PyRIT red teaming framework.

Professional Experience

Microsoft — Redmond, WA

AI Security Researcher II, AI Red Team — *Jan 2024–Present*

- Developed RL-based methods for training adversarial models, including (1) *GRP-Obliteration*, demonstrating that open-weight models can be broadly unaligned using a single prompt, and (2) *AdvGRPO*, a co-training method to automatically discover jailbreaks while hardening target model defenses.
- Designed Microsoft's first method for detecting poisoning attacks against open-weight language models, leveraging training data memorization and internal model signals to reconstruct backdoor triggers.
- Led red teaming of frontier AI systems (Microsoft and OpenAI), identifying high severity failure modes and driving iterative break-fix cycles.
- Active contributor to PyRIT, developing attack scenarios (e.g., GCG suffix generation, memorized training data extraction) and evaluation tooling to assess the reliability of LLM-based scoring systems.

Data & Applied Scientist, Azure — *Aug 2022–Dec 2023*

- Built an embedding-based system for clustering production incidents and detecting failures at scale.
- Developed pipelines to detect and prioritize kernel-mode memory leaks across the Azure fleet.

Selected Publications

B Bullwinkel, E Kim, A Minnich, M Russinovich. *Learning to Attack and Defend: Adaptive Red Teaming of Language Models via GRPO*. Preprint (2026).

B Bullwinkel, G Severi, K Hines, A Minnich, R S Siva Kumar, Y Zunger. *The Trigger in the Haystack: Extracting and Reconstructing LLM Backdoor Triggers*. Preprint (2026).

M Russinovich, Y Cai, K Hines, G Severi, **B Bullwinkel**, A Salem. *GRP-Obliteration: Unaligning LLMs With a Single Unlabeled Prompt*. Preprint (2026).

B. Bullwinkel et al. *Steering Language Model Refusal with Sparse Autoencoders*. ICML Workshop on Actionable Interpretability (2025).

B. Bullwinkel et al. *A Systematization of Security Vulnerabilities in Computer Use Agents*. ICML Workshop on Data in Generative Models (2025).

Education

Harvard University — Cambridge, MA
M.S. in Data Science — *2022*
GPA: 3.95/4.0; Thesis: *Generative Adversarial Network Methods for Solving Differential Equations*

Williams College — Williamstown, MA
B.A. in Mathematics, Chinese (*cum laude*) — *2020*
GPA: 3.83/4.0; Williams-Exeter Programme at Oxford (WEPO)

Technical Skills

Programming	Python, SQL, KQL
ML / AI	PyTorch, Hugging Face Transformers, TRL (RLHF/GRPO), vLLM, Ollama
Systems	Ray, DeepSpeed, Weights & Biases, Linux, Azure, AWS, Docker
Language	Mandarin (working proficiency)